



UNIVERSITY INTERSCHOLASTIC LEAGUE

Mathematics

Invitational A • 2026



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1. Keri's Kennel will board your dog when you are out of town. They charge a basic fee of \$29.95 per day, plus \$5.95 per meal for premium dog food, and \$39.95 per hour to exercise your dog. Find the cost to leave a dog with Keri for two weeks (14 days) if you want your dog to receive two meals each day and you want your dog to be exercised 30 minutes each day.

- (A) \$861.11 (B) \$862.22 (C) \$863.33 (D) \$864.44 (E) \$865.55

2. Joe is hoping to purchase a new RAV4 XLE Hybrid, which has an MSRP of \$36,958. If he agrees to the MSRP price and makes a down payment of \$7,500, the dealership is offering to finance the remainder of the cost, interest free, over 72 months. What will be the amount of each monthly payment? (nearest dollar)

- (A) \$409 (B) \$412 (C) \$415 (D) \$418 (E) \$421

3. Caleb has a jar full of nickels, dimes and quarters. The jar has a total of 312 coins with a value of \$44.35. There are 50 more quarters than dimes in the jar. How many nickels are in the jar?

- (A) 108 (B) 110 (C) 112 (D) 114 (E) 116

4-5. Consider line $\overleftrightarrow{AB}$ with points $A(-6, 8)$ and $B(4, -10)$.

4. The point $P(-11, b)$ lies on $\overleftrightarrow{AB}$. $b = \underline{\hspace{2cm}}$.

- (A) 13 (B) 15 (C) 17 (D) 19 (E) 21

5. The point $Q(a, 4)$ lies on the perpendicular bisector of $\overleftrightarrow{AB}$. $a = \underline{\hspace{2cm}}$.

- (A) 5 (B) 6 (C) 7 (D) 8 (E) 9

6-8. Consider a triangle with vertices $H(-8, 2)$, $I(2, 8)$, and $J(6, 6)$.

6. Find the perimeter of $\triangle HIJ$. (nearest tenth)

- (A) 30.5 (B) 30.7 (C) 30.9 (D) 31.1 (E) 31.3

7. Find the area of $\triangle HIJ$. (nearest whole number)

- (A) 22 (B) 23 (C) 24 (D) 25 (E) 26

8. Find the length of the shortest median of $\triangle HIJ$. (nearest tenth)

- (A) 5.0 (B) 5.2 (C) 5.4 (D) 5.6 (E) 5.8

9. Maverick can mow and edge 4 lawns in 15 hours. Evan can mow and edge 2 lawns in 9 hours. If they work together, how long would it take them to mow and edge 6 lawns? (nearest minute)

- (A) 12 hr 7 min (B) 12 hr 10 min (C) 12 hr 13 min (D) 12 hr 16 min (E) 12 hr 19 min

10. If $h(x) = -x^2 + bx + c$, and has zeros at $x = -6$ and $x = 3$, then the maximum value of $h(x)$ is _____. (nearest tenth)

- (A) 19.5 (B) 19.7 (C) 19.9 (D) 20.1 (E) 20.3

11. Points A, B, C and D lie on circle O such that chord $\overline{AC}$ intersects chord $\overline{BD}$ at point E. If $AE = 26$, $BE = 18$, and $DE = 40$, then $CE =$ _____. (nearest tenth)

- (A) 27.5 (B) 27.7 (C) 27.9 (D) 28.1 (E) 28.3

12. If $f(x) = 2x^3 + 18x^2 + cx - 50$ and $f(-4) = -10$, then $f(2) =$ _____.

- (A) 92 (B) 94 (C) 96 (D) 98 (E) 100

13. If $s(x)$ is the slant asymptote of the function $f(x) = \frac{x^2 - 4x + 5}{x + 1}$, then $s(8) =$ _____.

- (A) 2 (B) 3 (C) 4 (D) 5 (E) 6

14-15. The base of a pyramid is a square with each side equal to 16 cm. The height is 12 cm.

14. The volume of the pyramid is _____. (nearest whole number)

- (A) 1020 cm^3 (B) 1022 cm^3 (C) 1024 cm^3 (D) 1026 cm^3 (E) 1028 cm^3

15. The total surface area of the pyramid is _____. (nearest whole number)

- (A) 706 cm^2 (B) 709 cm^2 (C) 712 cm^2 (D) 715 cm^2 (E) 718 cm^2

16. Find the domain of the function $f(x) = \frac{\sqrt{x^2 - 36}}{|x - 7|}$.

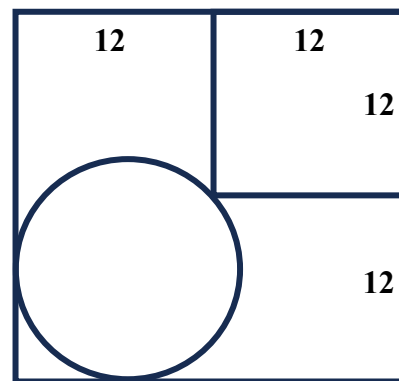
- (A) $(-\infty, -6] \cup [6, \infty)$ (B) $(-\infty, -6) \cup (6, 7) \cup (7, \infty)$ (C) $(-\infty, -6) \cup (6, \infty)$
(D) $(-\infty, \infty)$ (E) $(-\infty, -6] \cup [6, 7) \cup (7, \infty)$

17. Find the area of the circle. (nearest whole number)

- (A) 155 (B) 158 (C) 161
(D) 164 (E) 167

18. Find the number that is $\frac{3}{5}$ of the way from $-4\frac{1}{2}$ to $5\frac{1}{3}$.

- (A) $\frac{4}{5}$ (B) 1 (C) $\frac{6}{5}$ (D) $\frac{7}{5}$ (E) $\frac{8}{5}$



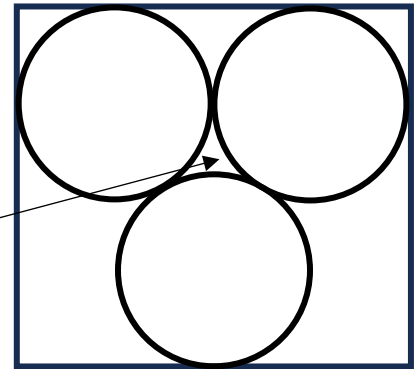
Problem 17

19-20. Three identical circles, each with a radius of 4.

19. Find the area of the rectangle. (nearest whole number)

- (A) 239
- (B) 241
- (C) 243
- (D) 245
- (E) 247

Region III



Problems 19, 20

20. Find the area of region III. (nearest hundredth)

- (A) 2.55
- (B) 2.58
- (C) 2.61
- (D) 2.64
- (E) 2.67

21. Given: $f(x) = 3x - 8$, $g(x) = 5x + 9$ and $h(x) = (f \circ g)(x)$. $h^{-1}(52) = \underline{\hspace{2cm}}$. (nearest tenth)

- (A) 2.2
- (B) 2.3
- (C) 2.4
- (D) 2.5
- (E) 2.6

22. A superball is dropped from a height of 16 feet. Each time it hits the floor, it bounces $\frac{9}{10}$ of its previous height. Find the total distance traveled by the ball. (nearest whole number)

- (A) 300 ft
- (B) 304 ft
- (C) 308 ft
- (D) 312 ft
- (E) 316 ft

23. Find the distance from the point $(2, 5, 7)$ to the plane $x + 3y - 2z = 12$. (nearest tenth)

- (A) 0.8
- (B) 1.2
- (C) 1.6
- (D) 2.0
- (E) 2.4

24. Four couples plan to attend a concert and sit in the same row. Each row has eight seats. If the two members of each couple are to sit together, how many different seating arrangements are possible?

- (A) 372
- (B) 376
- (C) 380
- (D) 384
- (E) 388

25. Three of the zeros of $f(x) = x^4 + bx^3 + cx^2 + dx - 20 = 0$ are $-5, 4, 2 + \sqrt{3}$. $f(2) = \underline{\hspace{2cm}}$.

- (A) 38
- (B) 40
- (C) 42
- (D) 44
- (E) 46

26. Consider the circle $x^2 + y^2 - 12x + 16y - 21 = 0$. The area of the circle is $\underline{\hspace{2cm}}$. (nearest whole number)

- (A) 380
- (B) 382
- (C) 384
- (D) 386
- (E) 388

27. Rotational latency is the delay waiting for the rotation of a hard drive disk to bring the required disk sector under the read-write head. It depends on the rotational speed of a disk measured in revolutions per minute. The average rotational latency is one-half the rotational period. If the rotational speed of the hard drive of my computer is 7200 rpm, then the average rotational latency of the hard drive disk is $\underline{\hspace{2cm}}$ milliseconds. (nearest hundredth)

- (A) 4.13
- (B) 4.17
- (C) 4.21
- (D) 4.25
- (E) 4.29

28. Find the period of the graph of $y = 2 - 3 \sin\left(\frac{\pi x}{6} + \frac{\pi}{4}\right)$.

- (A) 4 (B) 6 (C) 12 (D) π (E) 2π

29. The value of my Toyota Grand Highlander Hybrid depreciates exponentially. I originally paid \$61,355 on January 6, 2024. On January 6, 2026, the value had fallen to \$54,500. If the value continues to depreciate exponentially, then the expected value of my Highlander on January 6, 2036 is \$ _____. (nearest dollar)

- (A) 30,099 (B) 30,109 (C) 30,119 (D) 30,129 (E) 30,139

30. Find the shortest distance from the graph of $x^2 + y^2 = 56$ to the point $(10, -8)$. (nearest tenth)

- (A) 4.7 (B) 4.9 (C) 5.1 (D) 5.3 (E) 5.5

31-32. A regular octagon has a perimeter of 96 cm.

31. The area of the octagon is _____ cm^2 . (nearest whole number)

- (A) 686 (B) 689 (C) 692 (D) 695 (E) 698

32. The area of a circle circumscribed about the octagon is _____ cm^2 . (nearest whole number)

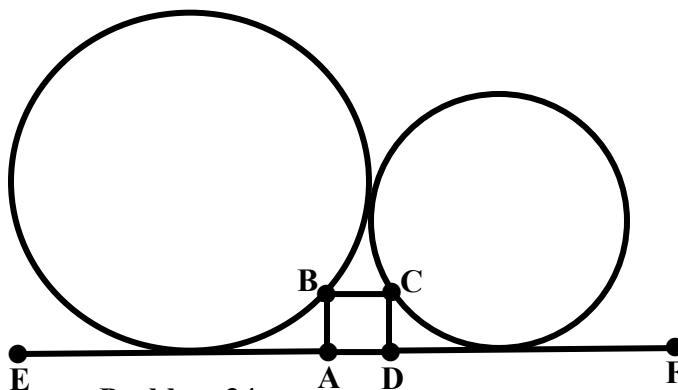
- (A) 769 (B) 772 (C) 775 (D) 778 (E) 781

33. The equation of an ellipse with vertices $(2, 0)$ and $(2, -8)$, and with a focal point at $(2, -1)$ is $16x^2 + by^2 + cx + dy + e = 0$. $e =$ _____.

- (A) 26 (B) 48 (C) 64 (D) 84 (E) 112

34. The circles are tangent to each other and to $\overline{EF}$. The area of the circles are 64π and 36π respectively. Find the area of square ABCD. (nearest hundredth)

- (A) 7.18
(B) 7.29
(C) 7.40
(D) 7.51
(E) 7.62



Problem 34

35. The y-intercept of the graph of the polar equation $r = \frac{-4}{3 \cos \theta - 5 \sin \theta}$ is $(0, b)$. $b =$ _____.
(nearest hundredth)

- (A) 0.80 (B) 0.82 (C) 0.84 (D) 0.86 (E) 0.88

36. The point $(a, -8)$ lies on the curve defined by the parametric equations $\begin{matrix} x = 3t^4 + 2t^2 \\ y = t^3 \end{matrix}$, $-6 \leq t \leq 6$.
 $a =$ _____.

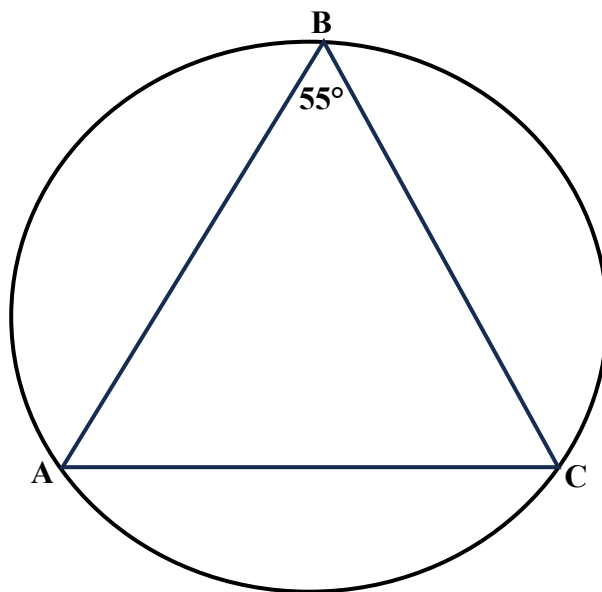
- (A) 50 (B) 52 (C) 54 (D) 56 (E) 58

37. Consider $\triangle ABC$ with $AB = BC$. The area of the circle is 94. Find the length of $\overline{AC}$.
(nearest hundredth)

- (A) 8.74
(B) 8.85
(C) 8.96
(D) 9.07
(E) 9.18

38. Consider a circle inscribed in $\triangle ABC$. Find the circumference of this circle.
(nearest tenth)

- (A) 16.8
(B) 17.1
(C) 17.4
(D) 17.7
(E) 18.0



Problems 37, 38

39. Find the eccentricity of the hyperbola $6x^2 - y^2 = 6$. (nearest tenth)

- (A) 1.8 (B) 2.0 (C) 2.2 (D) 2.4 (E) 2.6

40. Yogi, Clete and Joe are playing catch. The bearing from Yogi to Clete is 250° . The bearing from Joe to Clete is 340° . The bearing from Joe to Yogi is 30° . The distance from Clete to Joe is 120 feet. Find the distance from Joe to Yogi. (nearest whole number)

- (A) 187 ft (B) 190 ft (C) 193 ft (D) 196 ft (E) 199 ft

41. Consider the function $f(x) = -x^3 + 5x^2 - 4x - 3$. The sum of the values in the interval $(0.5, 6)$ that satisfy the mean value theorem for $f(x)$ on $[0.5, 6] =$ _____. (nearest tenth)

- (A) 3.7 (B) 3.9 (C) 4.1 (D) 4.3 (E) 4.5

42. A circle is inscribed in a square. The circumference of the circle is increasing at a constant rate of 8 inches per second. As the circle expands, the square expands to maintain the condition of tangency. At the instant the circle has an area of 36π square inches, the rate of increase in the area enclosed between the circle and the square is _____ square inches per second. (nearest tenth)

(A) 12.5 (B) 12.7 (C) 12.9 (D) 13.1 (E) 13.3

43. Find the average value of $f(x) = 3\sin(0.2x) + 2\cos(0.2x)$ on the interval $[6, 36]$. (nearest hundredth)

(A) -0.33 (B) -0.29 (C) -0.25 (D) -0.21 (E) -0.17

- 44-45. Let $y = f(x)$ be the solution to the differential equation $\frac{dy}{dx} = \frac{4x}{3y^2}$ with the initial conditions $0 < x < 12$ and $f(2) = 5$.

44. Use Euler's method to approximate $f(3)$ using two steps of equal size starting at $x = 2$. (nearest hundredth)

(A) 5.06 (B) 5.09 (C) 5.12 (D) 5.15 (E) 5.18

45. Find the exact value of $f(3)$. (nearest hundredth)

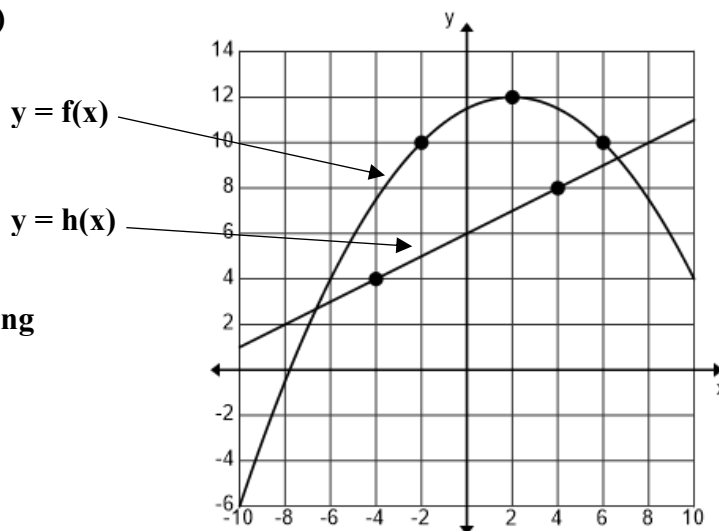
(A) 5.13 (B) 5.16 (C) 5.19 (D) 5.22 (E) 5.25

46. Find the area bounded by the graphs of $y = f(x)$ and $y = h(x)$. (nearest tenth)

(A) 48.3
(B) 48.6
(C) 48.9
(D) 49.2
(E) 49.5

47. Find the volume generated by revolving the region bounded by the graphs of $y = f(x)$ and $y = h(x)$ about the line $y = -2$. (nearest whole number)

(A) 3102
(B) 3106
(C) 3110
(D) 3114
(E) 3118



Problems 46, 47, 48

48. The points $P(-6, 4)$ and $Q(6, 10)$ lie on the curve $y = f(x)$. Find the arclength of arc PQ . (nearest tenth)

(A) 15.8 (B) 16.0 (C) 16.2 (D) 16.4 (E) 16.6

49. If $f''(x) = 36x^2 - 4$, $f(-1) = 3$, and $f(1) = 11$, then $f(-2) =$ _____.

- (A) 34 (B) 36 (C) 38 (D) 40 (E) 42

50. Given: $f(x) = \frac{1}{3}x^3 + x - 8$. Find the value of $(f^{-1})'(x)$ when $x = 4$.

- (A) $\frac{1}{10}$ (B) $\frac{1}{9}$ (C) $\frac{1}{8}$ (D) $\frac{1}{6}$ (E) $\frac{1}{4}$

51. Let $F(x)$ be an antiderivative of $\frac{x^3 + 4\sin(x)}{x^2 - 7}$. If $F(3) = 4.2$, then $F(5) =$ _____.

(nearest tenth)

- (A) 19.2 (B) 19.4 (C) 19.6 (D) 19.8 (E) 20.0

52. Given: $F(x) = \int_1^{3x} \sqrt{t^2 + 2t} \, dt$. The length of the curve $y = F(x)$ for $1 \leq x \leq 3$ is _____.

(nearest tenth)

- (A) 40.4 (B) 40.7 (C) 41.0 (D) 41.3 (E) 41.6

	1	2	3	4	5	6	7	8	9
Score	176	182	164	215	211	188	134	182	198

53-54. Azucena entered the Williamson County Bowling Marathon last week. Each contestant is required to bowl nine games. The table above shows the results.

53. What is the difference in the mean score and the median score, a positive number? (nearest tenth)

- (A) 1.0 (B) 1.3 (C) 1.6 (D) 1.9 (E) 2.2

54. Find the interquartile range of the scores.

- (A) 33 (B) 33.5 (C) 34 (D) 34.5 (E) 35

55. Assume that the mean length of an adult Sumatran rhinoceros is 11 ft 3 in with a standard deviation of 6 in. Coach Bowers owns a rhinoceros that is 12 ft long. This length places him at the _____ percentile.

- (A) 87th (B) 89th (C) 91st (D) 93rd (E) 95th

56. A researcher is testing the claim that 42% of Americans believe that inflation is the most important issue in America today. In a survey involving a simple random sample of 1000 Americans, 398 responded by placing inflation as their top concern. The researcher performed an appropriate test with $H_0 : p = 0.42$ and $H_a : p \neq 0.42$. He used an $\alpha = 0.05$ level as his standard. What was the value of the test statistic? (nearest hundredth)

- (A) -1.63 (B) -1.52 (C) -1.41 (D) -1.30 (E) -1.19

Score	1	2	3	4	5
Probability	.02	.06	.12	.15	.65

57. A large number of students from Highland Park take the BC Calculus exam each spring. Based on results over the last twelve years, Mr. Speir created the table above. Define X to be the AP score on the BC Calculus exam for a randomly selected student. Compute the mean of the random variable X . (nearest hundredth)

- (A) 4.29 (B) 4.31 (C) 4.33 (D) 4.35 (E) 4.37

Miles per Week	34	40	44	49	56	59
5K time (minutes)	19.4	18.5	18.3	17.9	17.2	16.9

Spencer keeps a record of his weekly mileage and his 5K times at cross country meets. Use the table above for problems 58 and 59.

58. Spencer plotted the data and observed a strong, negative, linear relationship between his weekly mileage and his 5K times. Statistical software generated a LSRL. Find the value of the residual for the data point (56 miles per week, 17.2 minutes). (nearest thousandth)

- (A) 0.017 (B) 0.019 (C) 0.021 (D) 0.023 (E) 0.025

59. Find the time predicted by the LSRL if Spencer increases his mileage to 72 miles per week. (nearest hundredth)

- (A) 15.44 min (B) 15.55 min (C) 15.66 min (D) 15.77 min (E) 15.88 min

60. Consider the difference D of two independent random variables X and Y . $D = X - Y$.

Given that $\sigma_X = 7.22$ and $\sigma_Y = 5.17$, what is σ_D , the standard deviation of D ?

(nearest hundredth)

- (A) 2.05 (B) 4.63 (C) 7.20 (D) 8.88 (E) 12.39

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**University Interscholastic League
MATHEMATICS CONTEST
HS • Invitational A • 2026
Answer Key**

1. E	21. A	41. B
2. A	22. B	42. D
3. C	23. E	43. E
4. C	24. D	44. C
5. D	25. C	45. A
6. B	26. A	46. B
7. A	27. B	47. E
8. A	28. C	48. D
9. D	29. E	49. C
10. E	30. D	50. A
11. B	31. D	51. B
12. D	32. B	52. E
13. B	33. C	53. B
14. C	34. E	54. D
15. E	35. A	55. D
16. E	36. D	56. C
17. A	37. C	57. D
18. D	38. B	58. B
19. A	39. E	59. C
20. B	40. A	60. D